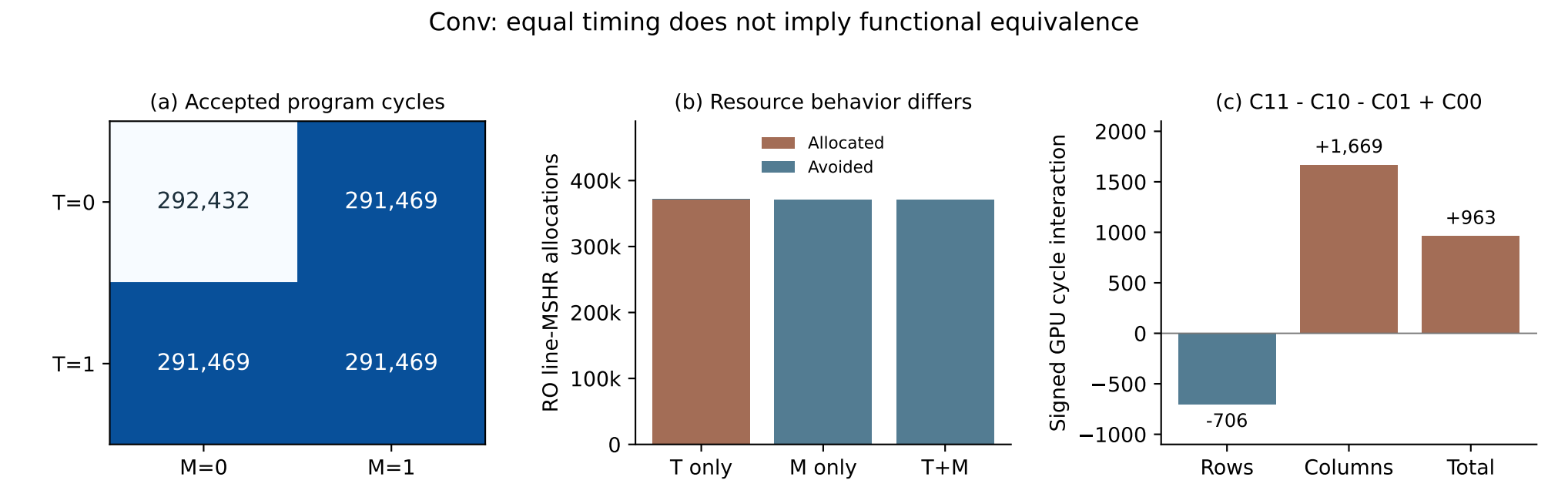
Conv: equal timing does not imply functional equivalence
(a) Accepted program cycles
T=0
T=1
292,432
291,469
291,469
291,469
M=0
M=1
(b) Resource behavior differs
RO line-MSHR allocations
Allocated
Avoided
400k
300k
200k
100k
0
T only
M only
T+M
(c) C11 - C10 - C01 + C00
Signed GPU cycle interaction
2000
1500
1000
500
0
−500
−1000
+1,669
+963
-706
Rows
Columns
Total
T or M enables the common RO path. Active corners have zero successful Tag detach.
Joint saving is 963 cycles, rather than the sum of the two either-on savings (1,926).